

Generalized Bloch Theorem for Spin Spirals in ABINIT

Le SHU

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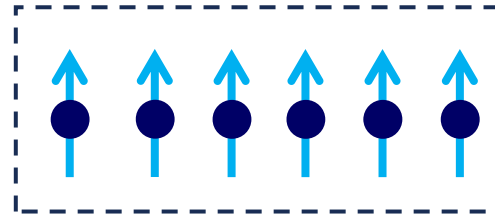
Université catholique de Louvain
Louvain-la-Neuve, Belgium
06/09/2026



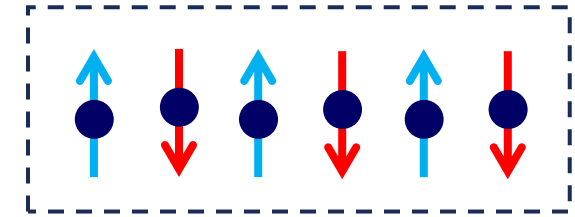
Magnetism

Collinear magnetic order

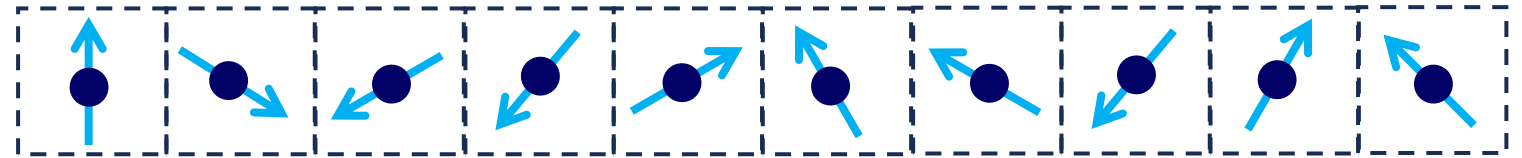
Ferromagnetic



Antiferromagnetic

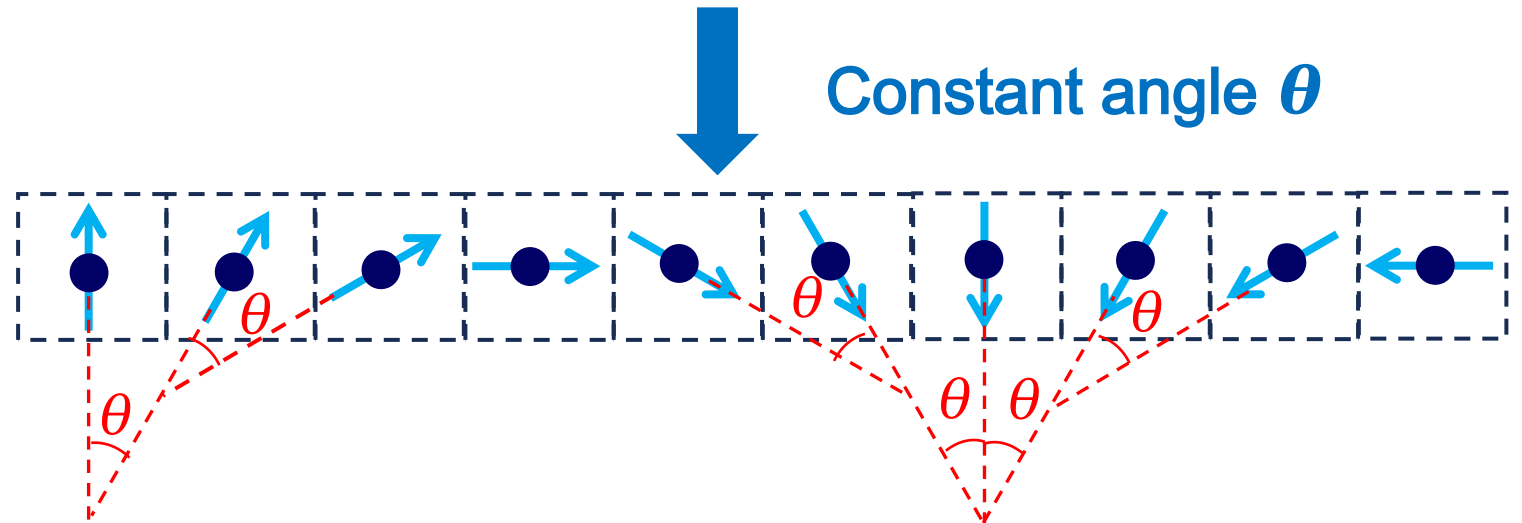


Noncollinear magnetic order



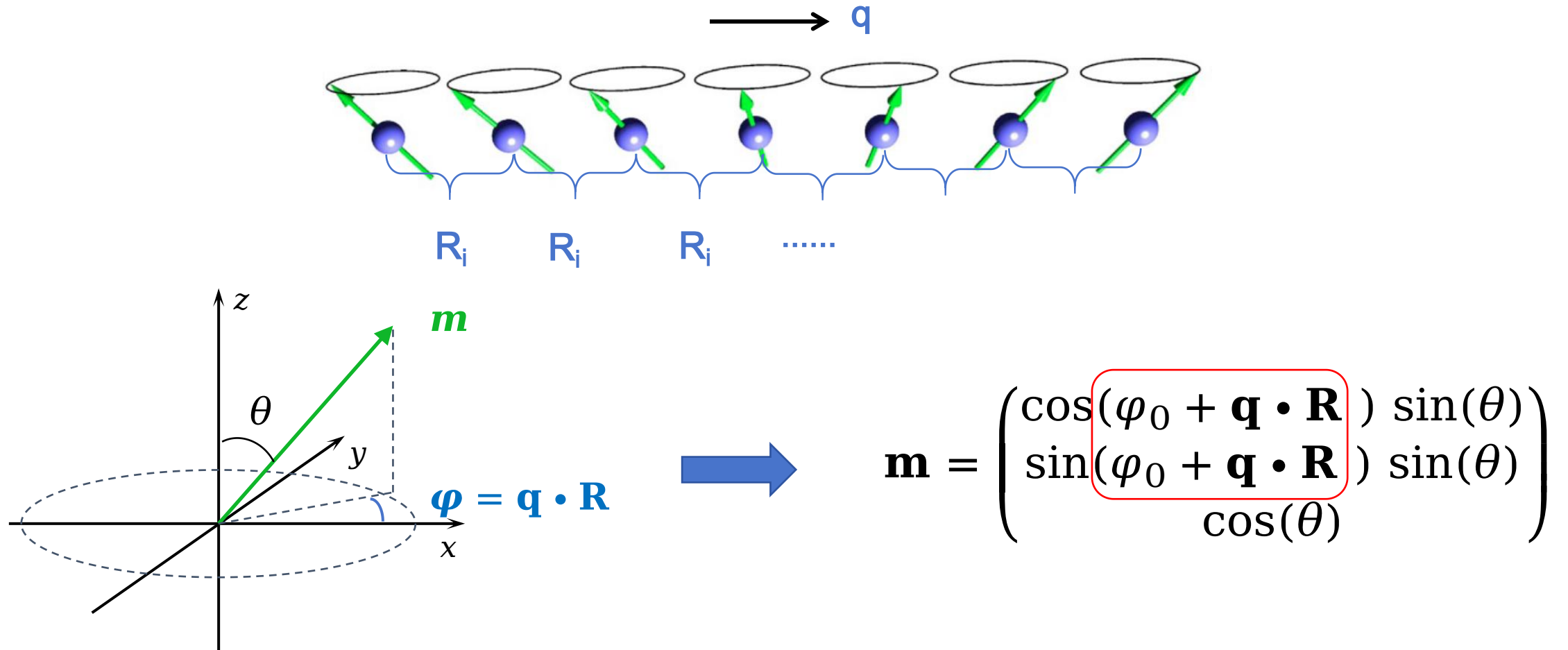
Constant angle θ

Spin spiral



What are spin spirals ?

(1) Propagation vector of spin spirals $\mathbf{q}$

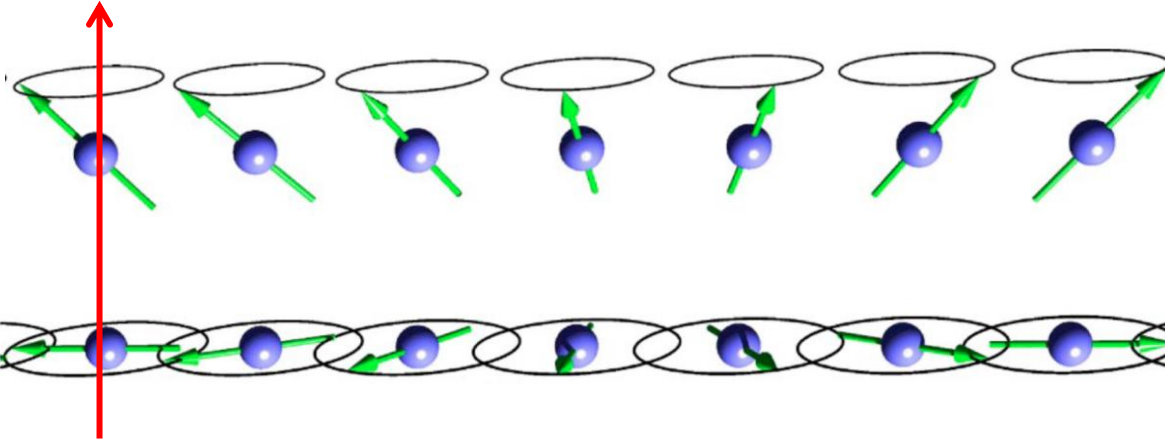


Kurz, et al. *Physical Review B* 69 (2004): 024415.

What are spin spirals ?

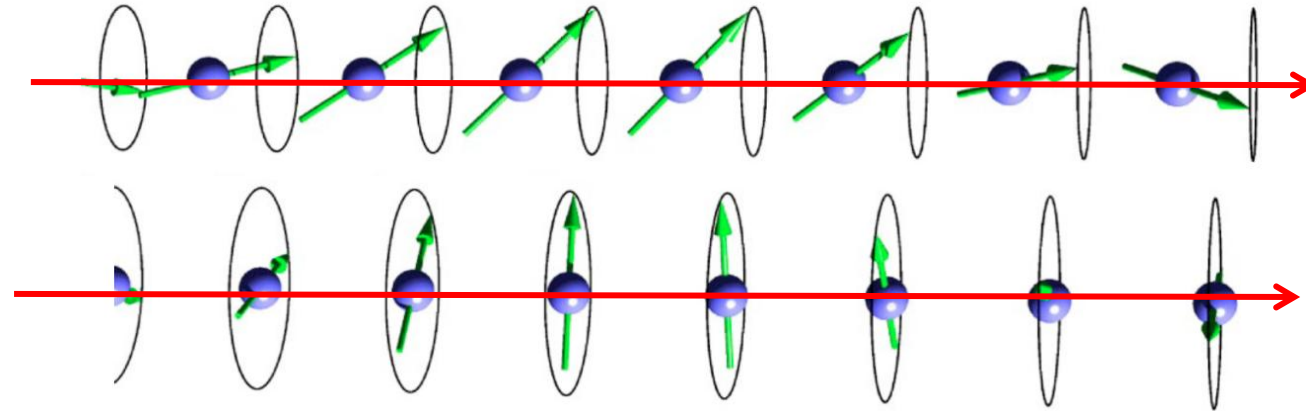
(2) Rotation axis

z-axis



$$\mathbf{m} = \begin{pmatrix} \cos(\varphi) \sin(\theta) \\ \sin(\varphi) \sin(\theta) \\ \cos(\theta) \end{pmatrix}$$

x-axis

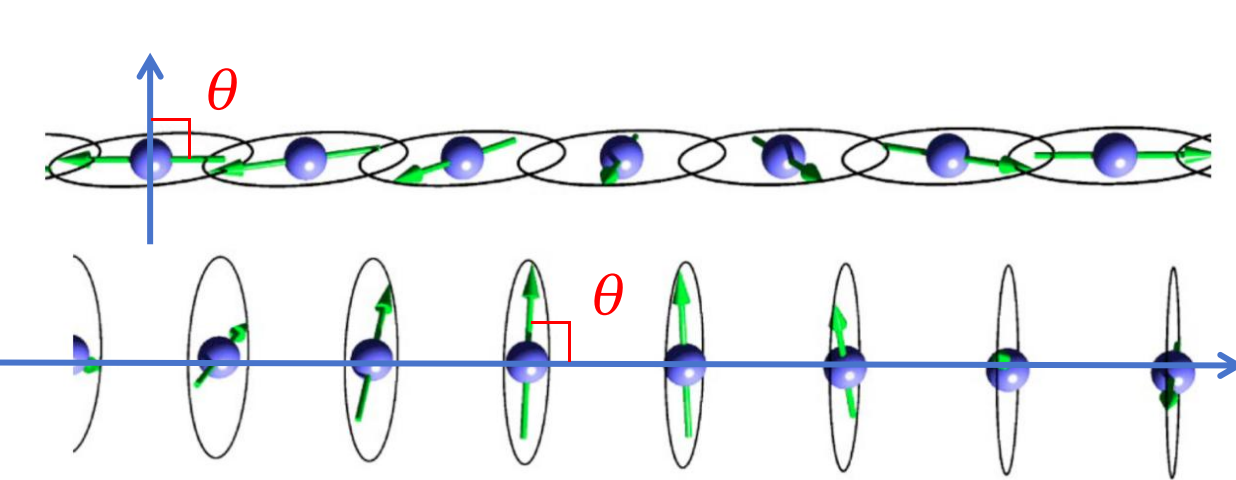


$$\mathbf{m} = \begin{pmatrix} \cos(\theta) \\ \cos(\varphi) \sin(\theta) \\ \sin(\varphi) \sin(\theta) \end{pmatrix}$$

Kurz, et al. *Physical Review B* 69 (2004): 024415.

What are spin spirals ?

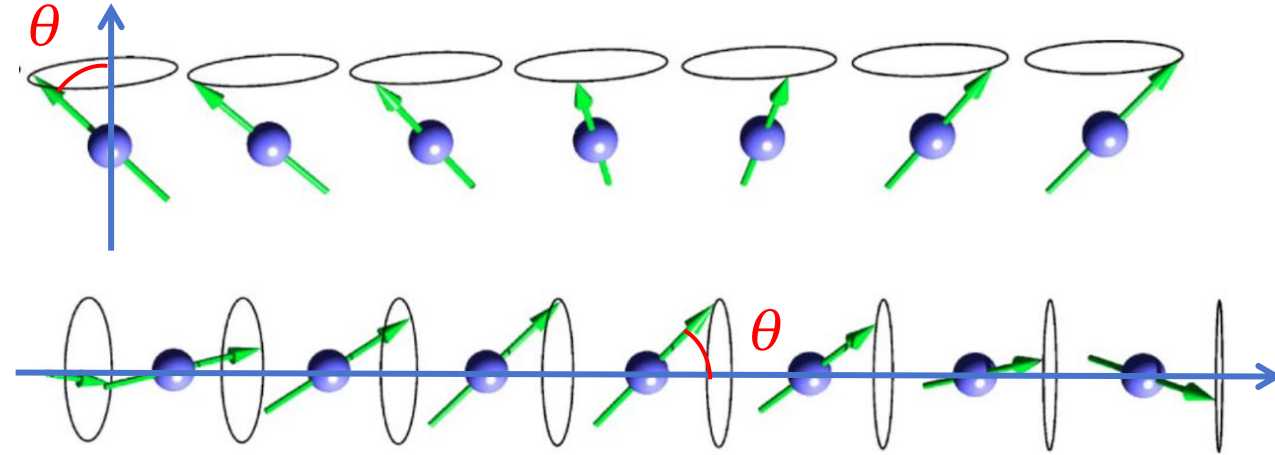
(3) Relative angle θ



Flat $\theta = \frac{\pi}{2}$



$$\mathbf{m} = \begin{pmatrix} \cos(\varphi_0 + \mathbf{q} \cdot \mathbf{R}) \\ \sin(\varphi_0 + \mathbf{q} \cdot \mathbf{R}) \\ 0 \end{pmatrix}$$



Conical $0 < \theta < \frac{\pi}{2}$



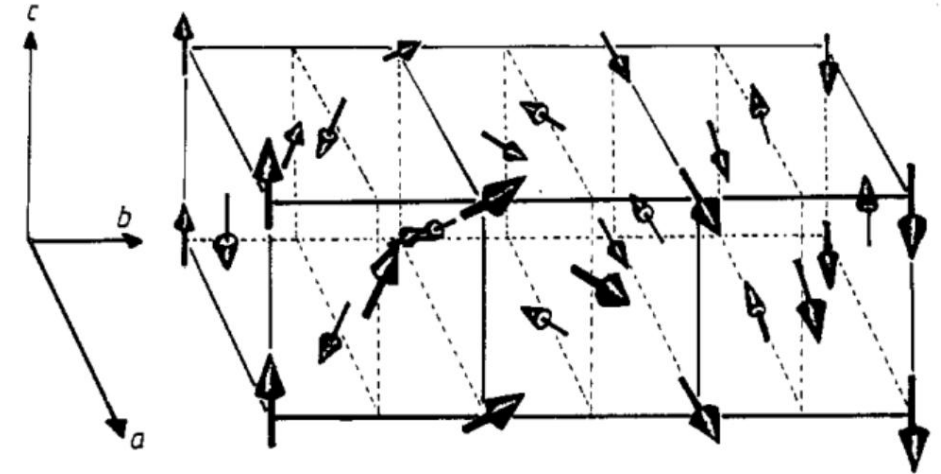
$$\mathbf{m} = \begin{pmatrix} \cos(\varphi_0 + \mathbf{q} \cdot \mathbf{R}) \sin(\theta) \\ \sin(\varphi_0 + \mathbf{q} \cdot \mathbf{R}) \sin(\theta) \\ \cos(\theta) \end{pmatrix}$$

Kurz, et al. *Physical Review B* 69 (2004): 024415.

Why spin spirals ?

Materials	T_c (K)	q (spiral vector)
FeP	125	(0, 0, 0.2)
$\text{Mn}_{0.7}\text{V}_{0.3}\text{As}$	142	(0.08, 0, 0)
$\text{Mn}_{0.65}\text{Cr}_{0.35}\text{As}$	195	(0.071, 0, 0)
γ - Fe	50	(0.1, 0, 1)
γ - $\text{Fe}_{9.7}\text{Co}_3$	30	(0.13, 0, 1)
CuO	220	(0.006, 0, 0.0017)
TbMnO_3	30	(4, 1/3, 1)
FeAs	70	(0, 0, 0.395)
CrAs	265	(0, 0, 0.356)

Several materials have ground-state spiral structure confirmed by experiments

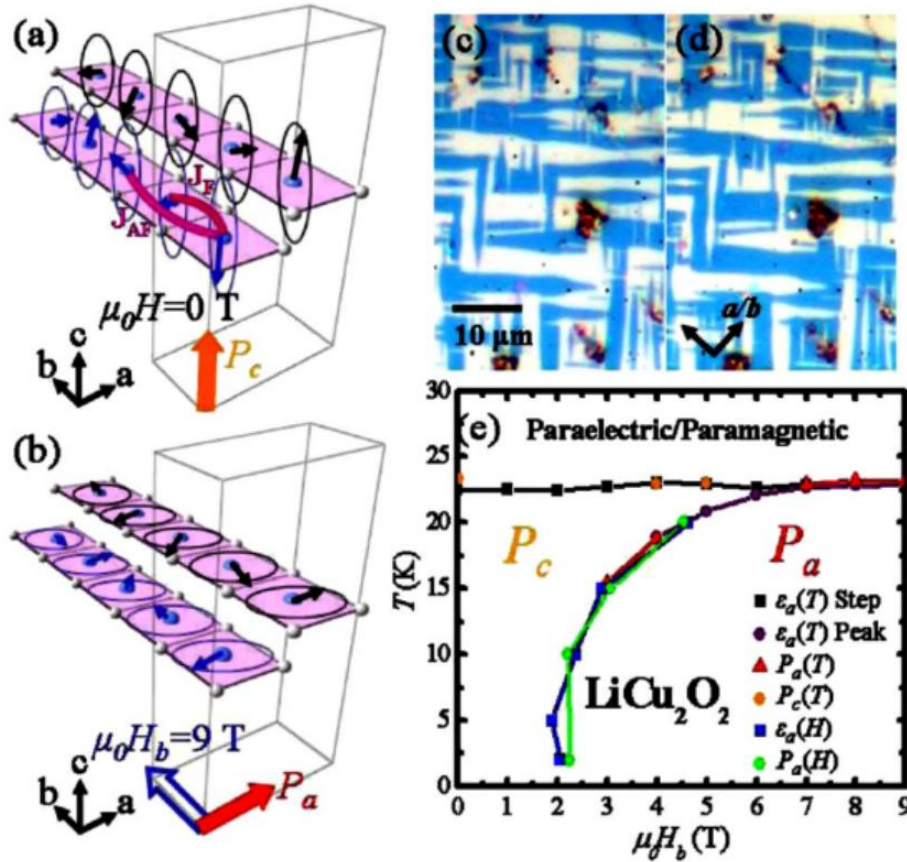


Spiral ground state in cubic γ -Fe by neutron scattering measurements

➤ **spin spiral ground state**

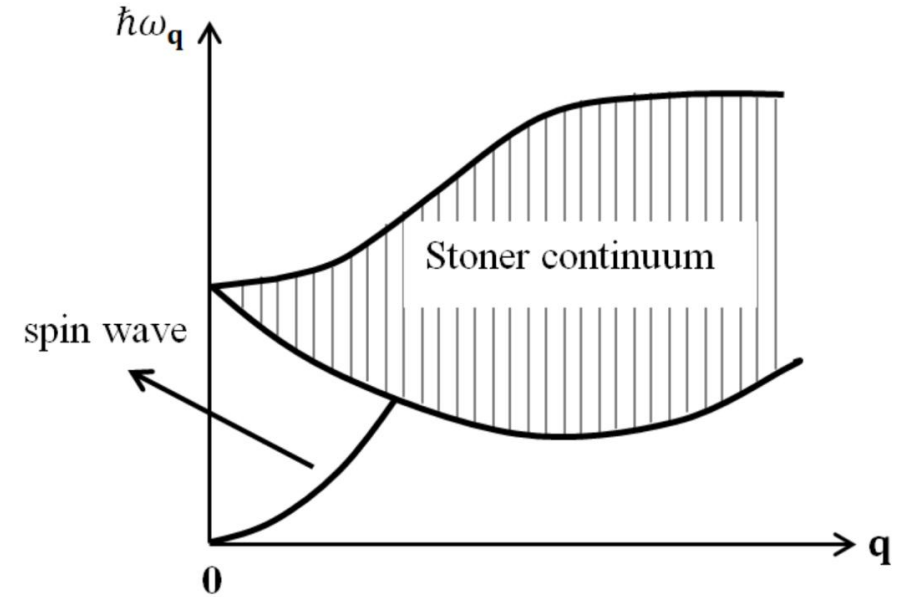
Y. Tsunoda, J. Phys: Condens. Matter 1 (1989): 10427.

Why spin spirals ?



Spin-spiral-driven ferroelectricity in LiCu_2O_2

- induce the ferroelectric property



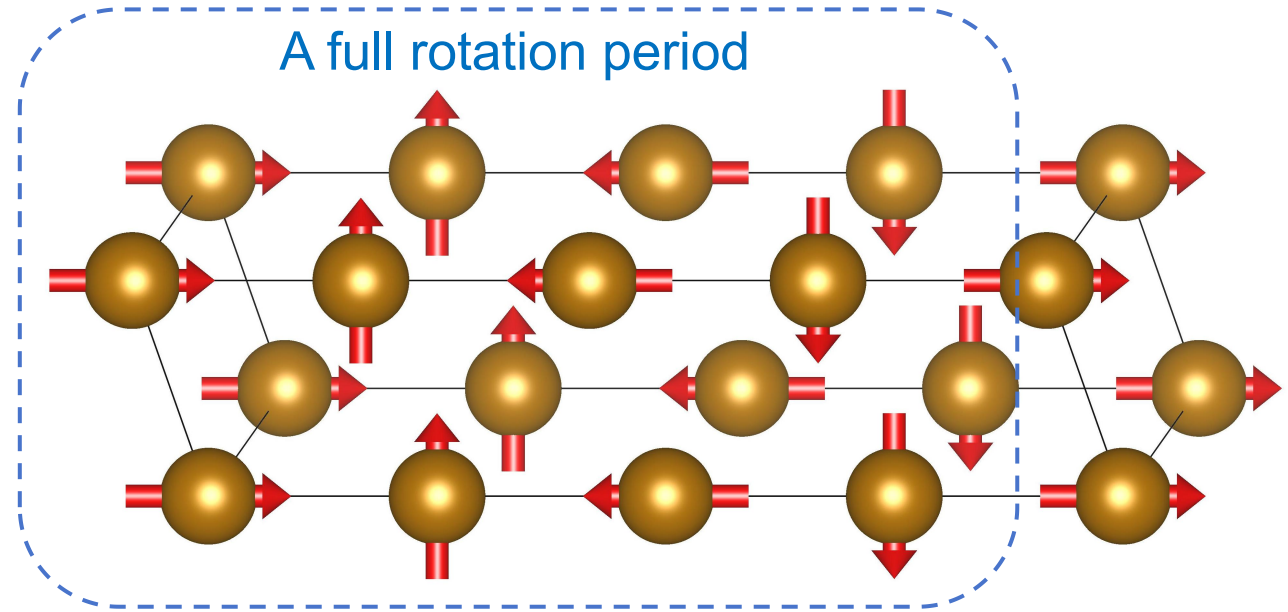
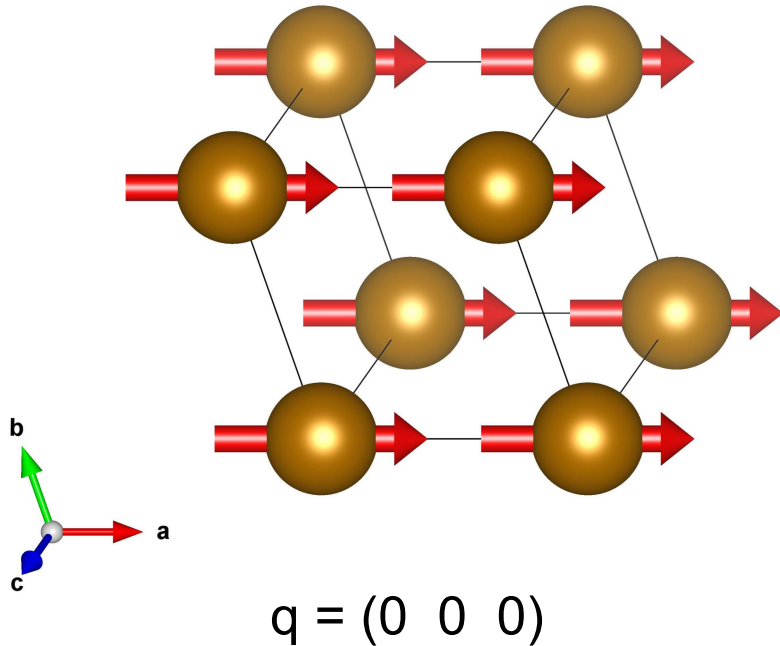
Spin waves below the Stoner continuum
long-wavelength spiral: extract D and magnon dispersion

- magnon dispersion $\hbar\omega(\mathbf{q})$
- calculate spin stiffness $\omega = Dq^2$

Park, S., et al. *Physical Review Letters* 98 (2007): 057601.

How to calculate spin spirals ?

➤ Supercell method



$q = (0.25 \ 0 \ 0)$ $\longrightarrow$ rotation angle $\varphi = 90^\circ$
supercell size: $4 \times 1 \times 1$

Small q
($q = 0.01$)



Very large supercell !

(need ~100 primitive cells) 



**Generalized Bloch Theorem
(GBT)**

Generalized Bloch Theorem

One-electron Hamiltonian:

$$\widehat{H} = -\frac{\hbar^2}{2m_e} \nabla^2 + V(\mathbf{r}) + \frac{\hbar}{2} \boldsymbol{\sigma} \cdot \mathbf{B}(\mathbf{r})$$

Exchange-correlation **B**-field for spin-spirals:

$$\mathbf{B}(\mathbf{r}) = B(\mathbf{r}) \begin{pmatrix} \cos(\varphi) \sin(\theta) \\ \sin(\varphi) \sin(\theta) \\ \cos(\theta) \end{pmatrix}$$

Rotation

$$\widehat{H}_{\mathbf{k}}^{\text{spin-spiral}} = \begin{pmatrix} \nabla^2 + V(\mathbf{r}) + B(\mathbf{r}) \cos(\theta) & B(\mathbf{r}) \sin(\theta) e^{-i\varphi} \\ B(\mathbf{r}) \sin(\theta) e^{i\varphi} & \nabla^2 + V(\mathbf{r}) - B(\mathbf{r}) \cos(\theta) \end{pmatrix}$$

Off-diagonal term:

$$\begin{pmatrix} \hat{H}_{\uparrow\uparrow}(\mathbf{k}, \mathbf{q}) & \hat{H}_{\uparrow\downarrow}(\mathbf{q}) \\ \hat{H}_{\downarrow\uparrow}(\mathbf{q}) & \hat{H}_{\downarrow\downarrow}(\mathbf{k}, \mathbf{q}) \end{pmatrix}$$

$$\widehat{H}_{\mathbf{k}}^{\uparrow\downarrow}(\mathbf{r} + \mathbf{R}) = \widehat{H}_{\mathbf{k}}^{\uparrow\downarrow}(\mathbf{r}) \cdot e^{-i\mathbf{q} \cdot \mathbf{R}}$$

$$\widehat{H}_{\mathbf{k}}^{\downarrow\uparrow}(\mathbf{r} + \mathbf{R}) = \widehat{H}_{\mathbf{k}}^{\downarrow\uparrow}(\mathbf{r}) \cdot e^{i\mathbf{q} \cdot \mathbf{R}}$$

Not periodic...

But Bloch-type phase !

Generalized Bloch Theorem

Unitarily transformed Hamiltonian:

$$\widehat{H}_{\mathbf{k},\mathbf{q}}^{\text{spin-spiral}} = U^\dagger \widehat{H} U \quad \downarrow \quad U = \begin{pmatrix} e^{i(\mathbf{k}-\frac{\mathbf{q}}{2})\cdot\mathbf{r}} & 0 \\ 0 & e^{i(\mathbf{k}+\frac{\mathbf{q}}{2})\cdot\mathbf{r}} \end{pmatrix}$$

$$\widehat{H}_{\mathbf{k},\mathbf{q}}^{\text{spin-spiral}} = \left(\begin{array}{c|c} ((\mathbf{k} - \frac{\mathbf{q}}{2}) - i\nabla)^2 + V(\mathbf{r}) + B_z(\mathbf{r}) & B(r) \sin(\theta) \exp(-i\varphi + i\mathbf{q} \cdot \mathbf{r}) \\ \hline B(\mathbf{r}) \sin(\theta) \exp(i\varphi - i\mathbf{q} \cdot \mathbf{r}) & ((\mathbf{k} + \frac{\mathbf{q}}{2}) - i\nabla)^2 + V(\mathbf{r}) + B_z(\mathbf{r}) \end{array} \right)$$

(periodic)

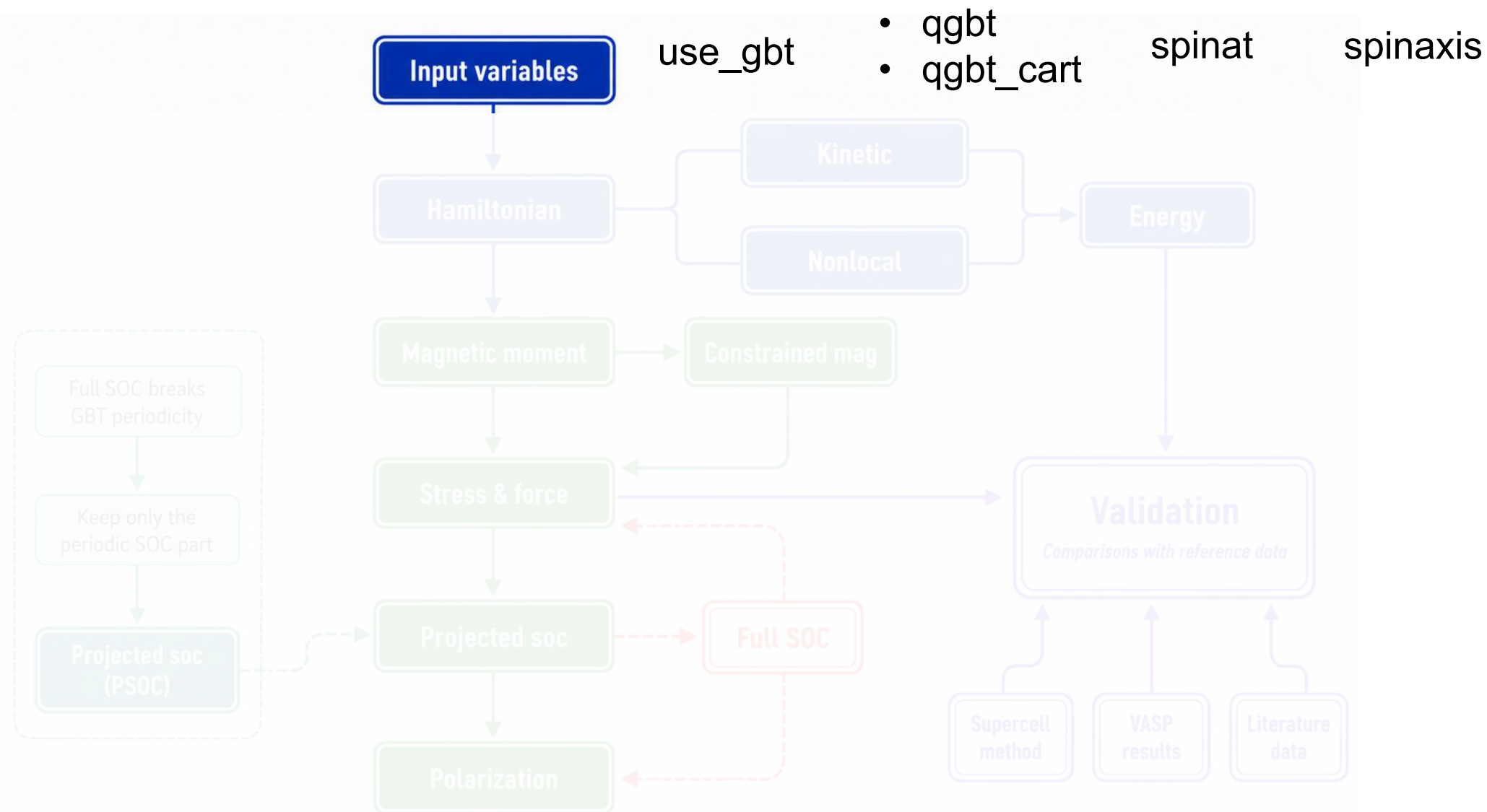
$$\downarrow \quad |\Psi\rangle \rightarrow |\Psi'\rangle = U^\dagger |\Psi\rangle$$

$$\Psi_{n\mathbf{k}}(\mathbf{r}|\mathbf{q}) = (N\Omega_0)^{-\frac{1}{2}} \begin{pmatrix} e^{i(\mathbf{k}-\frac{\mathbf{q}}{2})\cdot\mathbf{r}} u_{\mathbf{k}}^\uparrow(\mathbf{r}|\mathbf{q}) \\ e^{i(\mathbf{k}+\frac{\mathbf{q}}{2})\cdot\mathbf{r}} u_{\mathbf{k}}^\downarrow(\mathbf{r}|\mathbf{q}) \end{pmatrix} \rightarrow \begin{array}{l} \text{Generalized Bloch Theorem} \\ \text{Reduction to primitive cell !} \end{array}$$

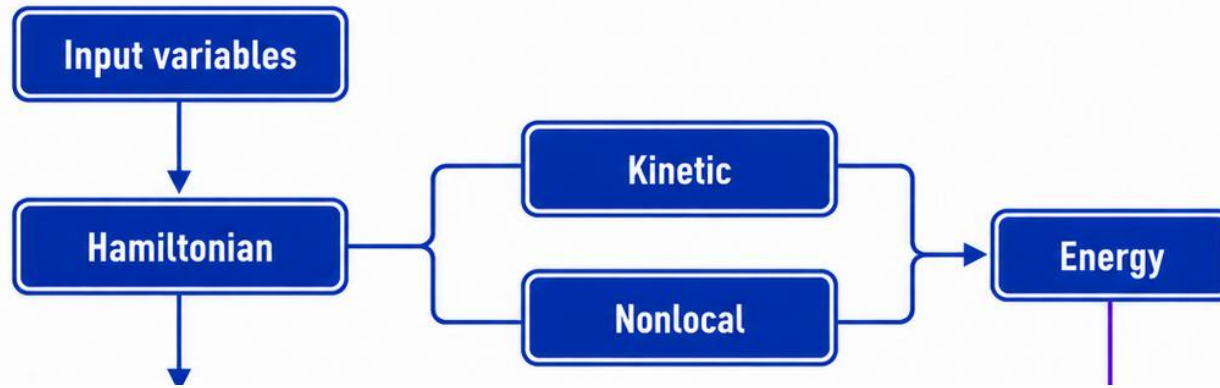
Phase factors for
spin rotation

Periodic functions

Steps in spin spiral Abinit implementation



Steps in spin spiral Abinit implementation



- Kinetic energy operator:

$$\widehat{T}_{\text{spin-spiral}}^{\uparrow\uparrow}(\mathbf{G}, \mathbf{G}') = \frac{1}{2} (\mathbf{k} - \frac{\mathbf{q}}{2} + \mathbf{G})^2 \delta_{\mathbf{G}, \mathbf{G}'}$$

$$\widehat{T}_{\text{spin-spiral}}^{\downarrow\downarrow}(\mathbf{G}, \mathbf{G}') = \frac{1}{2} (\mathbf{k} + \frac{\mathbf{q}}{2} + \mathbf{G})^2 \delta_{\mathbf{G}, \mathbf{G}'}$$

- ## plane wave basis

k

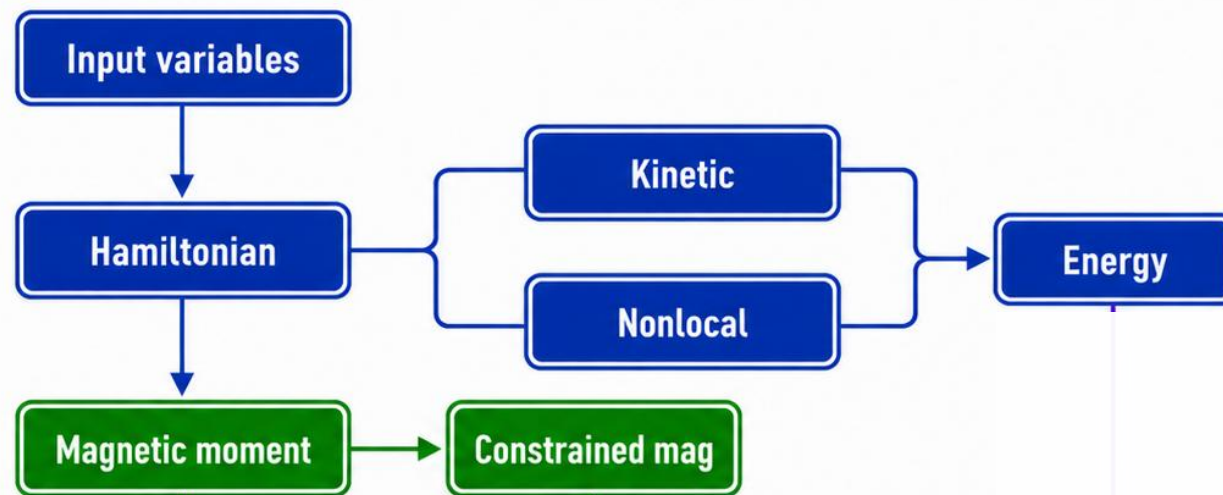


$$\mathbf{k} \pm \mathbf{q}/2$$

- **Nonlocal pseudopotential operator:**

$$\widehat{V}_{\text{spin-spiral}}^{\uparrow\uparrow}(\mathbf{G}, \mathbf{G}') = \frac{1}{\Omega_0} \sum_{\mu\kappa} E_{\mu\kappa} e^{-i(\mathbf{k} - \frac{\mathbf{q}}{2} + \mathbf{G}) \cdot \mathbf{R}_{\kappa}^a} \zeta_{\mu\kappa}(\mathbf{k} - \frac{\mathbf{q}}{2} + \mathbf{G}) \\ \times e^{i(\mathbf{k} - \frac{\mathbf{q}}{2} + \mathbf{G}') \cdot \mathbf{R}_{\kappa}^a} \zeta_{\mu\kappa}^*(\mathbf{k} - \frac{\mathbf{q}}{2} + \mathbf{G}')$$

Steps in spin spiral Abinit implementation



Spin density matrix:

$$\rho^{\alpha\beta}(\mathbf{r}) = \sum_n f_n \Psi_n^{\alpha}(\mathbf{r}) \Psi_n^{\beta*}(\mathbf{r})$$

**Periodicity breakdown
under GBT !**

$$\rho^{\uparrow\downarrow}(\mathbf{r} + \mathbf{R}) \neq \rho^{\uparrow\downarrow}(\mathbf{r})$$

$$\rho^{\downarrow\uparrow}(\mathbf{r} + \mathbf{R}) \neq \rho^{\downarrow\uparrow}(\mathbf{r})$$

Need to unspiral

$$\bar{\rho}^{\uparrow\downarrow}(\mathbf{r}) = e^{i\mathbf{q}\cdot\mathbf{r}} \rho^{\uparrow\downarrow}(\mathbf{r})$$

$$\bar{\rho}^{\downarrow\uparrow}(\mathbf{r}) = e^{-i\mathbf{q}\cdot\mathbf{r}} \rho^{\downarrow\uparrow}(\mathbf{r})$$

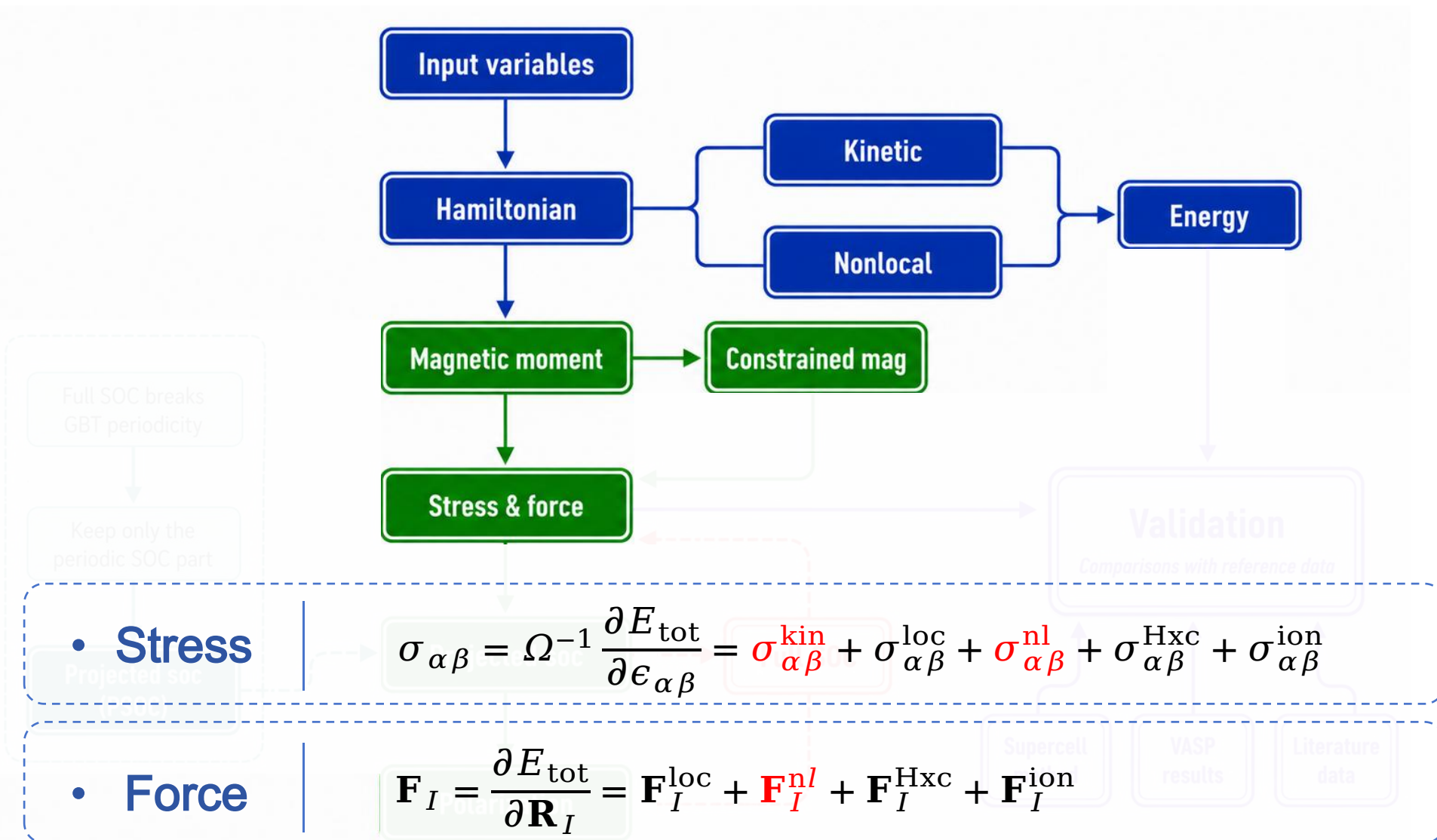
Magnetization:

$$m_z(\mathbf{r}) = \rho^{\uparrow\uparrow}(\mathbf{r}) - \rho^{\downarrow\downarrow}(\mathbf{r}),$$

$$m_x(\mathbf{r}) = e^{-i\mathbf{q}\cdot\mathbf{r}} \bar{\rho}^{\uparrow\downarrow}(\mathbf{r}) + e^{i\mathbf{q}\cdot\mathbf{r}} \bar{\rho}^{\downarrow\uparrow}(\mathbf{r}),$$

$$m_y(\mathbf{r}) = i(e^{-i\mathbf{q}\cdot\mathbf{r}} \bar{\rho}^{\uparrow\downarrow}(\mathbf{r}) - e^{i\mathbf{q}\cdot\mathbf{r}} \bar{\rho}^{\downarrow\uparrow}(\mathbf{r}))$$

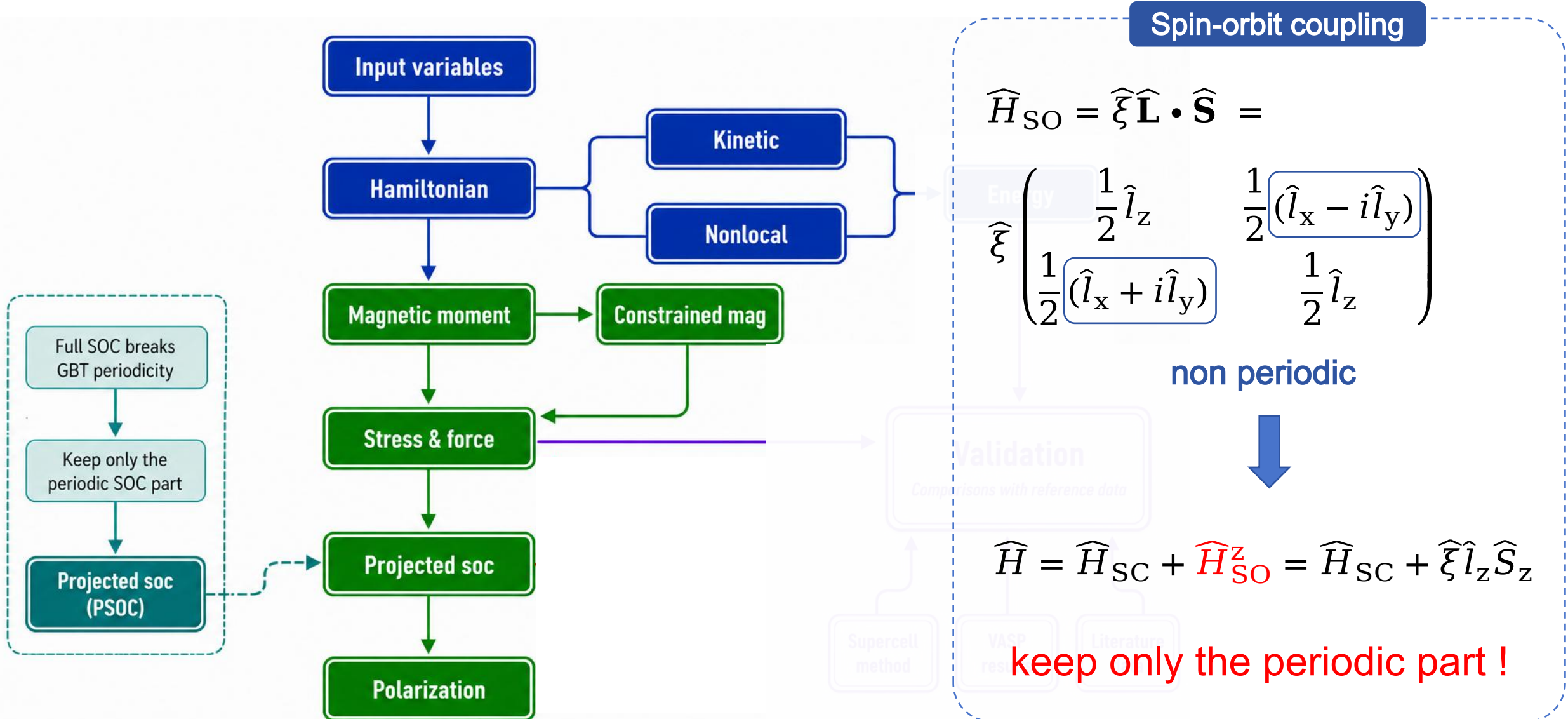
Steps in spin spiral Abinit implementation



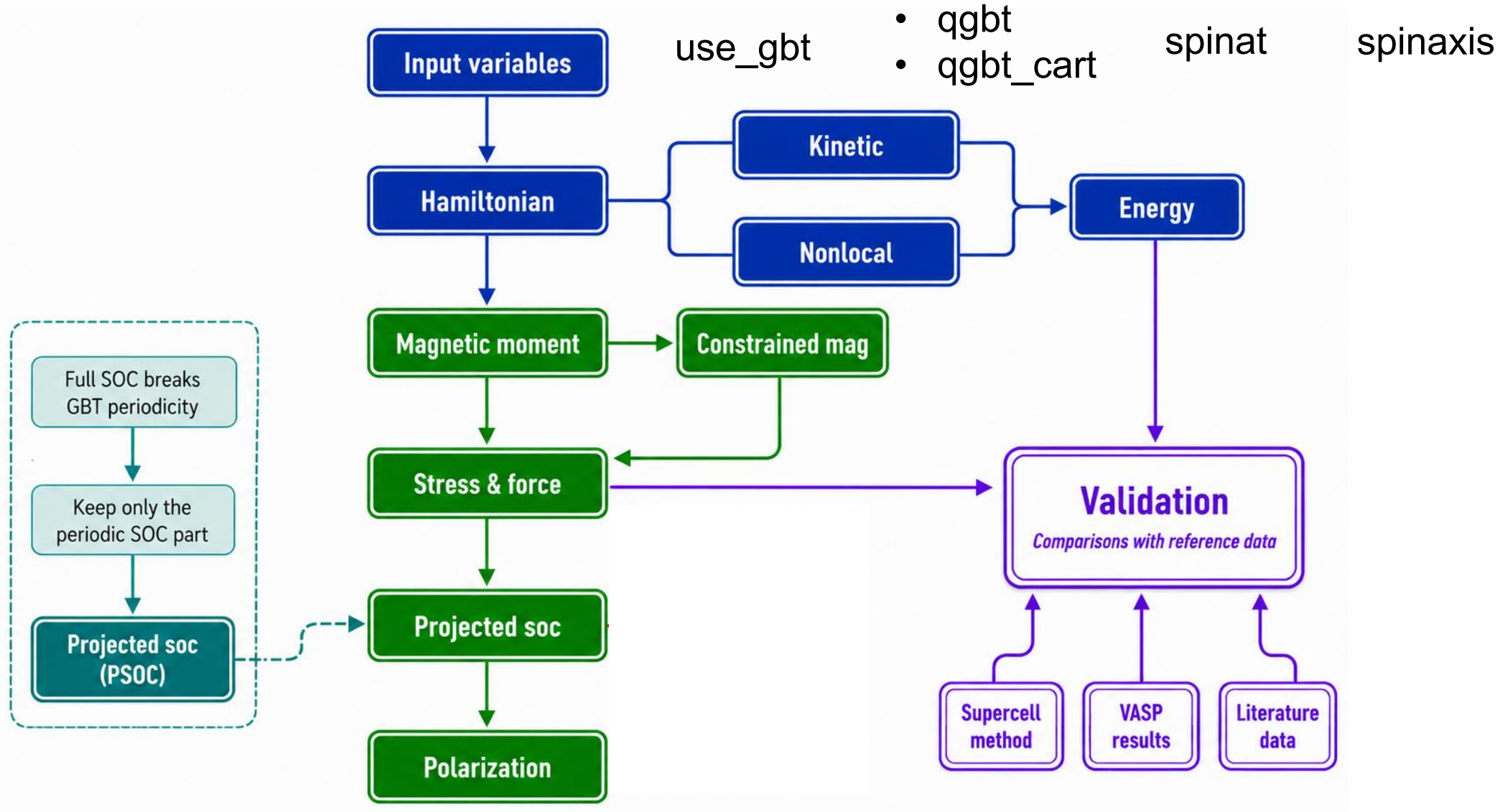
Plane-wave basis shift

$$\mathbf{k} \rightarrow \mathbf{k} \pm \mathbf{q}/2$$

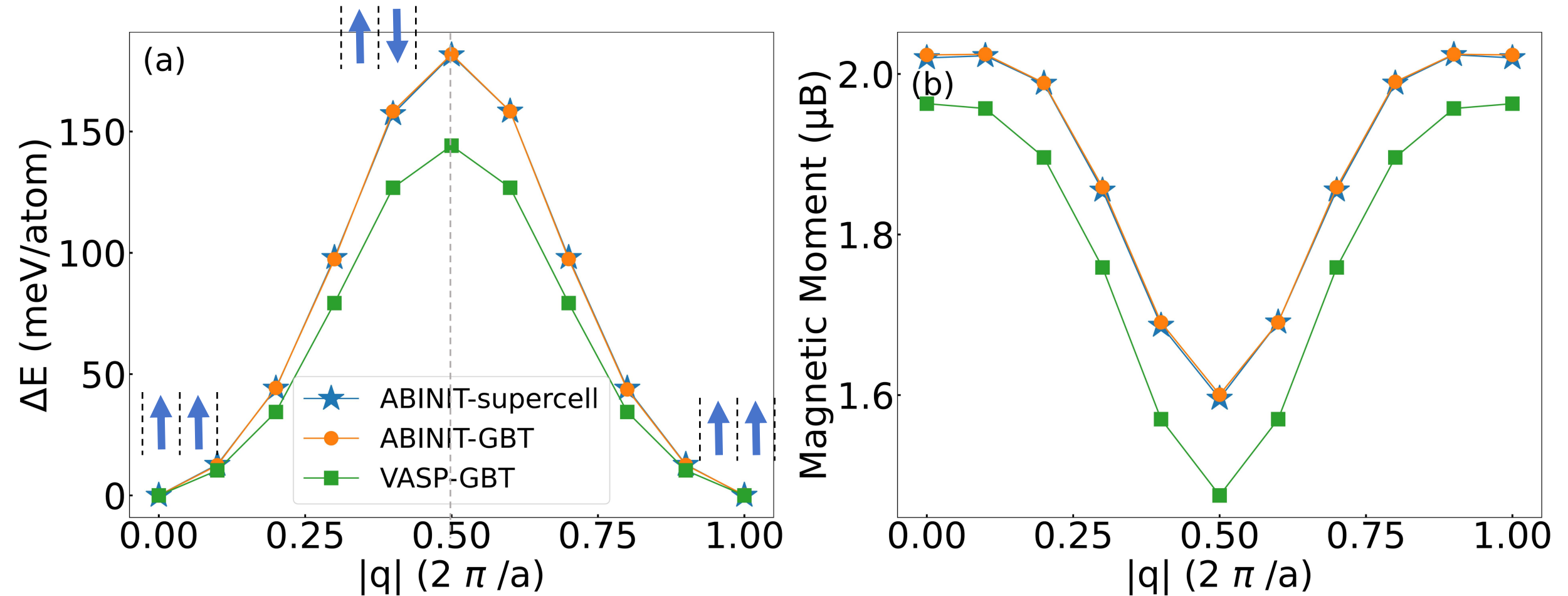
Steps in spin spiral Abinit implementation



Steps in spin spiral Abinit implementation



Total energy & Magnetic moment in BCC Fe

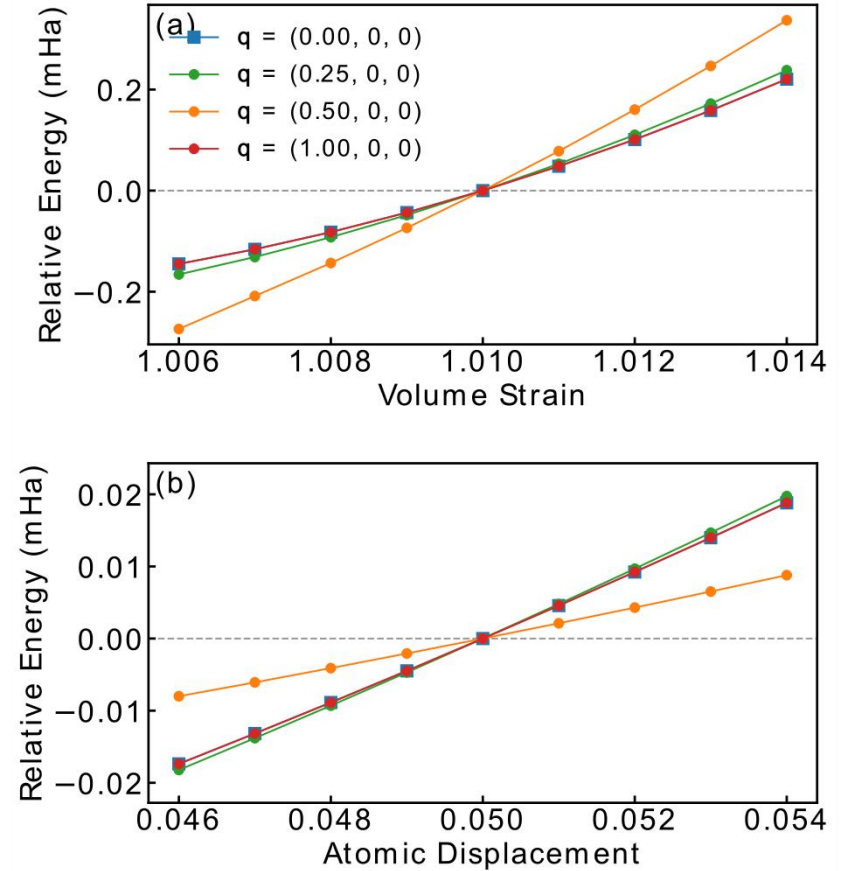


The three approaches yield consistent results

Stress & Force in BCC Fe

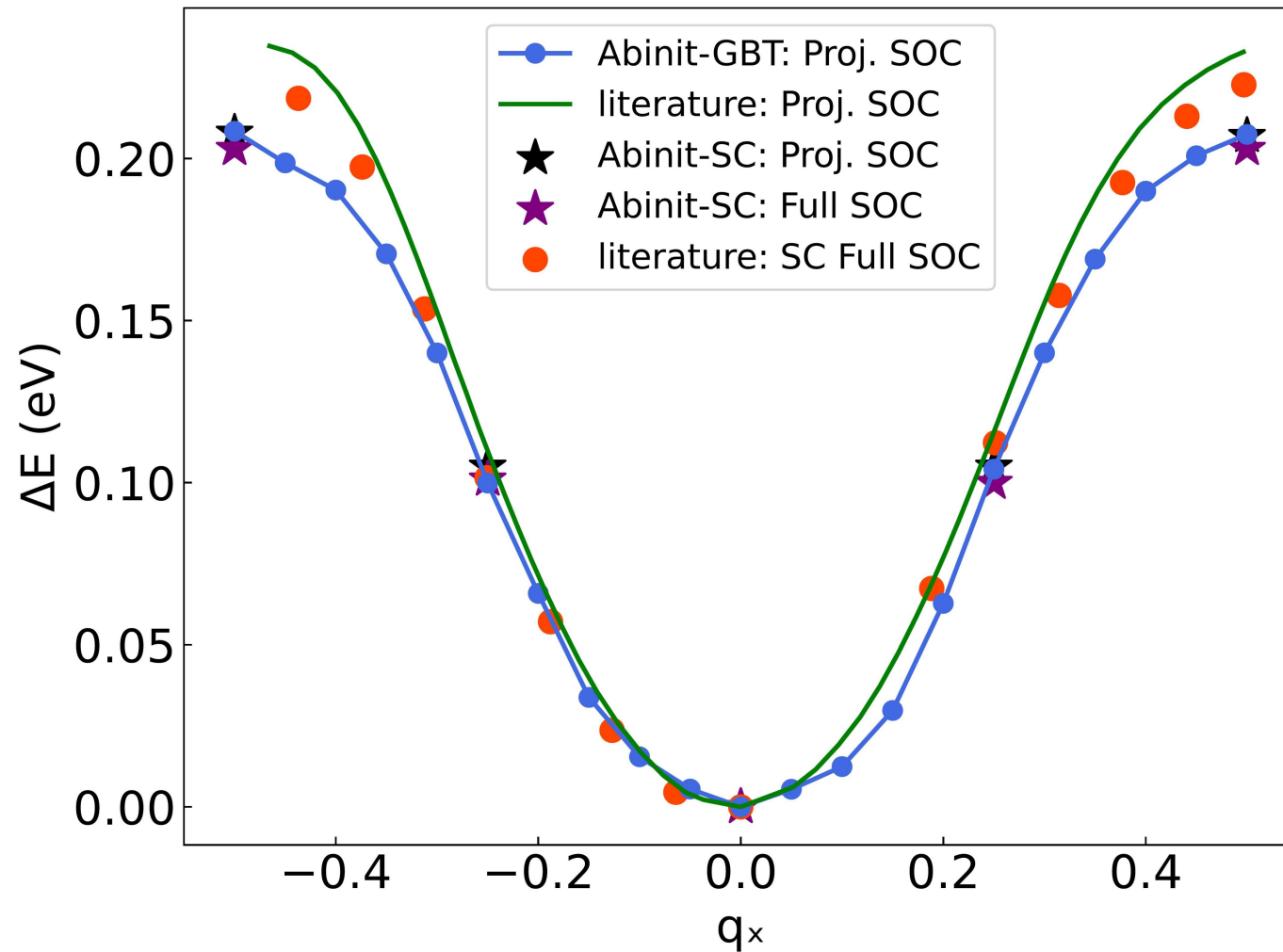
$q_{\text{gbt}} (\frac{2\pi}{a})$	Stress ($\times 10^{-4}$ Hartree/Bohr ³)			Forces ($\times 10^{-3}$ Hartree/Bohr)	
	ABINIT-GBT	GBT-FD	Supercell	ABINIT-GBT	GBT-FD
0 0 0	6.39	6.39	6.39	4.53	4.53
1 0 0	6.40	6.39	6.39	4.53	4.53
0.25 0 0	7.07	7.06	7.20	4.75	4.75
0.50 0 0	10.6	10.6	10.6	2.09	2.10

↔ agree
↔ agree
↔ agree



- Periodicity: $q_{\text{gbt}} = (0\ 0\ 0) \equiv q_{\text{gbt}} = (1\ 0\ 0)$,
- Excellent agreement among ABINIT-GBT, finite difference, and supercell.

Projected Spin-orbit Coupling in Co/Pt bilayer



PSOC captures the full-SOC trend and agrees with supercell PSOC.

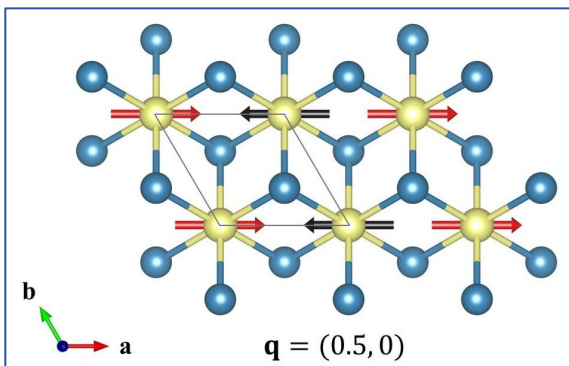
Electric Polarization in NiBr₂ monolayer

$q_x (\frac{2\pi}{a})$	Electronic polarization ($\mu\text{C}/\text{m}^2$)								
	GBT+PSOC			SC+PSOC			SC+Full SOC		
	P_x	P_y	P_z	P_x	P_y	P_z	P_x	P_y	P_z
0	0	0	0.06	0	0	0	0	0	0
0.25	59.53	-103.94	0.16	59.53	-103.73	0.07	57.02	-99.24	0.32
1/3	5.25	-9.22	0.01	5.24	-9.21	0.01	4.74	-8.37	0.03
0.5	0.20	-0.30	-0.10	0	0	0	0	0	0
1	0	0	0.10	0	0	0	0	0	0

agree

close

GBT+PSOC reproduces the spin-spiral-induced polarization.



Electric Polarization in NiBr₂ monolayer

gKNB model

$$\mathbf{P} = \sum_{ij} M^{ij} (S_i \times S_j)$$

1st-nearest neighbor

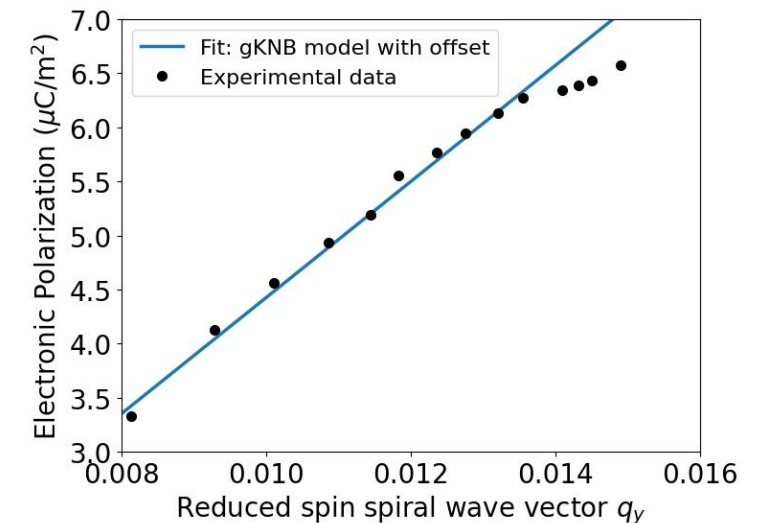
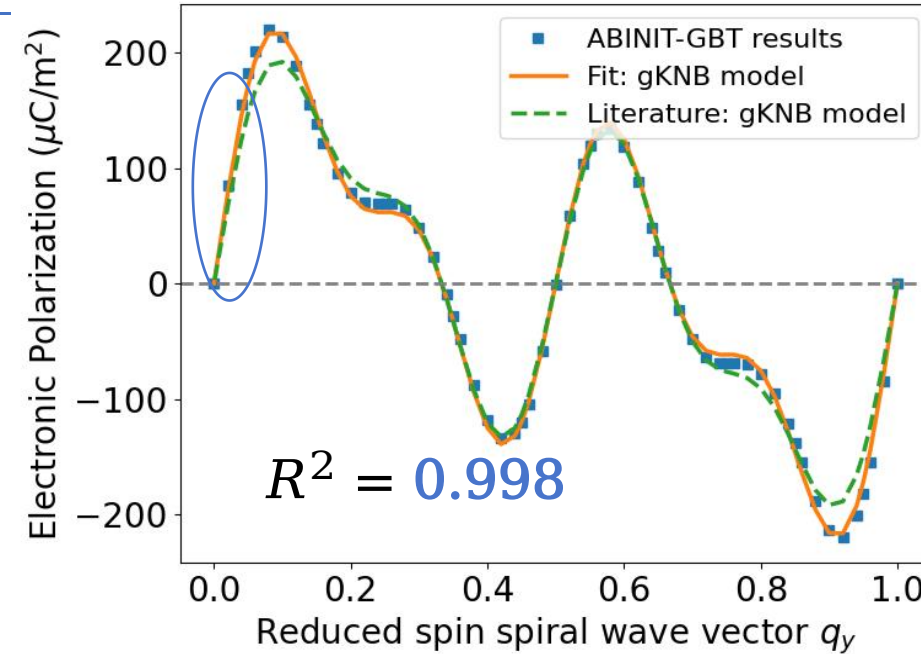
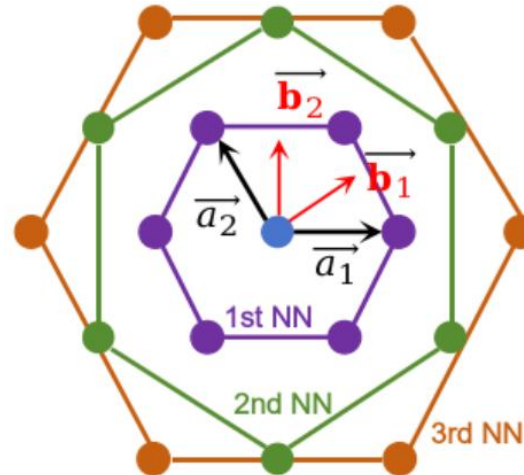


$$P_y(q) = 2[M_{23}^1(\sin(4\pi q) + \sin(2\pi q))$$

2nd- and 3rd-
nearest neighbor



$$P_y(q) = 2[M_{23}^1(\sin(4\pi q) + \sin(2\pi q)) + M_{13}^2 \sin(6\pi q) + M_{23}^3(\sin(8\pi q) + \sin(4\pi q))]$$



Summary and Outlook

➤ Summary

- ✓ Implemented a **self-consistent GBT in ABINIT** for primitive-cell spin spiral calculations.
- ✓ Enabled **key quantities**: total energies, magnetic moments, stress/forces, projected SOC and electric polarization.
- ✓ **Validated** all components against VASP, supercells, and finite-difference benchmarks, showing **excellent/close agreement**.

➤ Outlook

- Extend the framework to **full SOC** in spin-spiral calculations.
- Explore applications of spin spirals in **ferroelectric, magnetoelectric** materials.
- Apply the method to studies of **noncollinear magnetism** and **spin excitations**.

Thank you for your attention!

